



Ashmit Verma

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Education

Indraprastha Institute of Information Technology, Delhi
B.Tech (Computer Science and Design), Batch of 2028
2024 - present

CGPA: 7.00

Arwachin International School, Dilshad Garden, Delhi
CBSE, Standard 12, PCM+CS
2023-2024

Percentage : 85.4%

Arwachin International School, Dilshad Garden, Delhi
CBSE, Standard 10
2021 - 2022

Percentage : 90.4%

Skills

Expertise Area:	Data Structures and Algorithms, Software Design, Backend Development, Database Design and Query Optimisation, NoSQL and Document Data Modelling, Vector Search, Multi-Tenant Systems, Agentic Systems, Retrieval-Augmented Generation (RAG), Linux Server Management
Tech Languages:	C++ (Main), C, Python, Java, JavaScript, SQL
Tools and Technologies:	MongoDB, MongoDB Atlas, Atlas Vector Search, MySQL, PostgreSQL, Google Cloud SQL, Qdrant, BM25, XGBoost, LangChain, LangGraph, FastAPI, Flask, Node.js, REST APIs, Git, Linux, Vercel, React, JavaFX, Power BI, Adobe Illustrator, Photoshop, Fusion 360
Technical Electives:	Edge AI in Microcontrollers, (Udemy) DSA (C/C++), Full-Stack Web Development, Python Pro Bootcamp

Work Experience

Foqal CareOS

(May 2026 – Present)

Software Engineer Intern | *Foqal Analytics*

CareOS: an AI-powered Clinical Decision Support System for hospital ward monitoring, discharge and billing.

- Drove a billing-critical failure to **zero** — quoted minimums falling below money already charged, previously **32.4% of estimates** — by reframing the model's target so the constraint held **by construction**, not by post-hoc clamping.
- Sharpened forecasts **4x across a stay** (**105.7% MAPE at admission** → **24.2% by day 10**), by engineering cumulative clinical-usage features over **MIMIC-IV EHR data (7,077+ cardiology admissions, 46K+ day-wise records)**.
- Beat the production baseline by **3.1 MAPE points (44.5% → 41.3%)** on held-out patients, by benchmarking **14 model variants** and re-architecting the winner as **XGBoost quantile regression** over *remaining*, not total, cost.
- Gave nurses **one bedside view of deteriorating patients**, by building a real-time **NEWS2** Early Warning System (**Vite, JavaScript, FastAPI**) with drug-lab interaction alerts, escalation workflows and CCU step-down eligibility.
- Made LLM-generated **NABH-compliant discharge summaries** legally defensible, via audit trails and digital physician sign-off, and built into the backend **RAG** pipeline over **MedCPT** embeddings in **Qdrant**.

- Made a **4-source knowledge base** queryable in natural language, by building a **RAG** pipeline on **MongoDB Atlas** modelling transcripts, chunks and embeddings as **single documents** — no join at query time — with Groq-hosted **Llama 3.1 8B**.
- Turned **raw meeting URLs into knowledge** with no manual upload, by engineering a real-time ingestion pipeline integrating a teammate's **ML video-processing API** for **keyframe-level transcript extraction**, normalised alongside live **Slack history, PDFs and links** into one corpus.
- Guaranteed **no user can retrieve another tenant's data**, by enforcing isolation at the **Atlas Vector Search** index itself — **\$vectorSearch pre-filters** on user and date backed by **compound indexes** — not post-retrieval filtering.
- Lifted recall on ambiguous questions while **stopping invented answers**, by layering **BM25 over vector results** with overlap re-ranking, then gating every response on **lexical/entailment overlap** via an LLM verifier that **abstains instead of guessing**.
- Took the system to production in Slack, by building a **FastAPI** server with signature verification, **idempotent-upsert** ingestion, **MongoDB aggregation pipelines** for session history and a **9-stage LangGraph state machine** routing each query, deployed serverlessly on Vercel.
- Made the agent **proactive, not reactive**, by shipping a cron-scheduled supervisor that rebuilds an **evolving goal document** on every ingestion and auto-delivers Slack check-ins from goals, recent meetings and prior Q&A — over **multi-tenant** storage with persistent chat memory.

Projects

University ERP System ([GitHub](#))

Role-based ERP desktop app in **Java (JavaFX)** with Student/Instructor/Admin dashboards on a secure **two-database architecture** (Auth + ERP DB) — bcrypt-hashed passwords, RBAC, referential constraints, modular service layer, JUnit-style testing and CSV/PDF export.

Team Size-2

(Sep, 25 – Nov, 25)

Raashan – Supply Chain Management System ([GitHub](#))

Normalized **MySQL** DB modelling end-to-end supply flow — **transactions**, triggers, stored procedures, indexed batch tracking, automated low-stock reordering and analytical views; **ACID compliance** and consistent inventory state under **concurrent updates**.

Team Size-3

(Jan, 26 – Apr, 26)

Custom ISA Assembler & Simulator ([GitHub](#))

Python two-pass assembler and execution simulator for a custom ISA — symbol-table resolution, binary encoding, cycle-level tracing and an automated test harness validating against reference traces.

Team Size-3

(Jan, 25 – Mar, 25)

RAD.YO – Retro Digital Radio ([GitHub](#))

Retro digital radio web app in **React** with live streaming APIs, station switching and **hand-crafted UI assets designed in Adobe Illustrator/Photoshop (OSDC Hackathon Winner, Jul 2025)**.

Team Size-3

(Jul, 25)

Awards and Achievements

- **Codeforces Specialist** (1400+) | **250+ DSA problems** in **C++** (Codeforces, LeetCode, TUF, HackerRank)
- Ranked **26th, Goldman Sachs India Hackathon** | **Top 15,000** globally, **Google Code Jam Big Code**
- **OSDC Hackathon Winner 2025** – 1st Prize (RAD.YO) + CTF Winner (DSA) ([Hackathon](#), [CTF](#))

Positions of Responsibility

- Events Lead – Odyssey '25, IIIT Delhi ([Unstop](#)) | Operations Lead – TechSpectra, IIIT Delhi
- Events Lead – ACM Student Chapter (SIGCHI), IIIT Delhi (ashmitv@acm.org)

Interests and Hobbies

- Football · 3D CAD Modelling · UI/UX Design

Declaration: The above information is correct to the best of my knowledge.